

Abstract

Quantum coherence in photosynthetic light-harvesting complexes is well documented, but coherence has yet to be engineered as a design resource in an operating device. Here we do so by coupling a 9-site Fenna–Matthews–Olson (FMO) complex to a nanoparticle-on-mirror (NPoM) plasmonic nanocavity and a semi-transparent organic photovoltaic (OPV) layer through a 9×9 dressed Hamiltonian, propagated using the numerically exact process-tensor hierarchy of pure states (PT-HOPS/SBD; $L = 8$, $K = 2$, $n = 100$). The device operates as an agrivoltaic digital twin. Under dual-band (750 nm and 820 nm) spectral filtering, the forward transfer yield increases from 0.71 ± 0.02 to 0.89 ± 0.03 and exciton coherence extends by 50 % to 420 ± 35 fs in the 99 % passive fraction, in line with our companion study [1]. Confining active NPoM coupling to a localized 1 % sentinel fraction preserves an area-weighted global trapping yield of 97.1 % while enabling in-situ surface-enhanced Raman scattering (SERS) diagnostics. An FAO-56 Penman–Monteith model under the spectrally selective shield reduces crop evapotranspiration by 28 %. BB84 quantum key distribution over buried fiber secures IoT telemetry at quantum bit error rate QBER = 4.8 %, below the unconditional-security threshold. A cooperative deployment scenario in Cameroon yields a 4.32 yr payback, establishing engineered photonic environments and single-molecule plasmonic readout as operational quantum-technology elements outside the laboratory.

Keywords: non-Markovian quantum dynamics, quantum biology, plasmonic nanosensing, quantum agrivoltaics, digital twin, BB84 quantum cryptography

1 Introduction

Two-dimensional electronic spectroscopy has established long-lived coherence in the Fenna–Matthews–Olson (FMO) complex, and vibronic coupling is now understood to govern its excitation energy flow; yet in essentially all studies the photonic environment is treated as a fixed noise source to be tolerated, not a degree of freedom to be designed [2, 3, 4]. This gap matters wherever photosynthesis and photovoltaics must share light: agrivoltaics co-locates PV systems and crops to mitigate land-use competition [5, 6], but conventional semi-transparent PV claddings act as static filters that attenuate irradiance uniformly, trading electrical output for crop biomass without optimizing the spectral light-sharing window [7, 8]. The photosynthetic apparatus of green plants and bacteria has evolved under a spectrally structured photon flux, where specific wavelengths drive charge separation while others dissipate as heat [9]. This spectral specificity allows the PV layer to transmit only the photosynthetically active wavelengths while absorbing the remainder, converting the food-energy conflict into a shared spectral budget.

In green sulfur bacteria, the FMO complex shuttles excitation from the chlorosomal antenna to the reaction center [10]; its seven bacteriochlorophyll *a* sites show oscillatory coherence signatures that persist for hundreds of femtoseconds at cryogenic [2] and room temperature [3]. Although the functional relevance of these quantum beats remains debated, consensus has emerged that vibronic coupling — the hybridization of electronic

transitions with underdamped protein vibrations — governs the flow of excitation energy [4, 11]. Non-Markovian frameworks such as the hierarchical equations of motion [12] and the process tensor hierarchy of pure states (PT-HOPS) [13, 14] capture these effects exactly.

Previous work demonstrated that spectral filtering of the excitation pulse — aligning its spectrum with underdamped vibronic resonances at 750 nm and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO complex [1]. This selective vibronic excitation prepares polaron-dressed states that decouple from the dissipative protein-solvent bath. Single-photon experiments have confirmed that quantum coherence persists in individual light-harvesting complexes under ambient conditions [15], and quantum phase synchronization via exciton-vibrational energy dissipation has been identified as a mechanism sustaining long-lived coherence [16]. Here we extend this principle in three directions.

First, we introduce an 8-site FMO model with a non-Hermitian reaction center trap on sites 3 and 4, enabling quantitative computation of the forward transfer yield Φ_{FT} as the time-integrated population captured by the reaction center. Second, we embed the FMO complex in a nanoparticle-on-mirror (NPoM) plasmonic cavity [17] that couples the biological light-harvesting system to the organic photovoltaic (OPV) layer through strong coherent coupling, forming a 9×9 dressed Hamiltonian. Third, we incorporate dynamic photo-protection mechanisms — Floquet Stark detuning and optomechanically induced transparency (OMIT) — that mimic the non-photochemical quenching used by plants under high light stress.

The central question of this work is whether strong exciton-plasmon hybridization — normally a decoherence channel — can be confined to enable single-molecule quantum sensing without compromising global excitation transport, and whether numerically exact non-Markovian dynamics can quantify this design trade-off end to end. We map this microscopic optimization onto a multi-scale lifecycle assessment, parameterizing crop evapotranspiration with the FAO-56 model [18] under the OPV shield, computing the Net Ecological Benefit (NEB) across three scenarios, and evaluating cooperative ownership models. To prevent malicious manipulation of food-energy telemetry and protect agricultural data sovereignty, we secure the IoT network using BB84 quantum key distribution [19]. This quantum-to-organism perspective [20] bridges angstrom-scale vibronic couplings with square-meter agricultural yields, providing a unified framework for symbiotic quantum agrivoltaics (Fig. 4).

2 Results

2.1 8-Site FMO Hamiltonian with reaction center trap

Table 1: **8-site FMO site energies and key electronic couplings (cm^{-1}).** Full coupling matrix (28 unique pairs) provided in [Section S1](#).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
ε_n	280	420	0	110	270	500	310	200
$J_{n,n+1}$	—	−87.7	30.8	−53.5	−70.7	81.1	39.7	12.0

We construct an 8-site model of the FMO complex, extending the standard 7-site

framework [21] by including a coupling to the eighth bacteriochlorophyll that bridges the baseplate to the reaction center (Site 8 $\sim 200 \text{ cm}^{-1}$ above Site 3). The electronic Hamiltonian is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (1)$$

where ε_n are the site energies (relative to Site 3), J_{nm} are the inter-site electronic couplings (all 28 unique pairs defined in Section S1), and $\Gamma_{\text{RC}} = 0.15 \text{ fs}^{-1}$ is the effective irreversible reaction-center trapping rate, corresponding to a capture time $\tau_{\text{trap}} \approx 6.7 \text{ fs}$ — short compared with the 1-ps propagation window, so that near-complete capture (the 0.98 passive baseline) is dynamically accessible in Eq. (4) [22]. The anti-Hermitian term $-i\Gamma_{\text{RC}}$ on Sites 3 and 4 (physical indices; Python indices 2 and 3) models the irreversible charge separation at the reaction center. All 28 inter-site couplings are validated to be finite, and the Hamiltonian is verified to be non-Hermitian only on the trapping diagonal terms (Section S1).

2.2 NPoM plasmonic strong coupling

The OPV-FMO interface is mediated by a nanoparticle-on-mirror (NPoM) architecture [17], where a gold nanoparticle (diameter 80 nm) is separated from a gold mirror by a 1 nm graphene monolayer spacer. The graphene barrier prevents Dexter-type electron transfer while maintaining electromagnetic field enhancement. The NPoM cavity supports an intense localized surface plasmon mode that couples coherently to the FMO optical transitions and mediates the optical interface with the OPV layer.

The single-emitter coupling strength follows the cavity quantum electrodynamics scaling:

$$g_0 = g_{\text{ref}} \sqrt{\frac{V_{\text{ref}}}{V_{\text{mode}}}}, \quad (2)$$

with $g_{\text{ref}} = 120 \text{ cm}^{-1}$ at $V_{\text{ref}} = 1 \text{ nm}^3$ [17]. At the sub-nanometer mode volumes accessible in NPoM cavities ($V_{\text{mode}} < 1 \text{ nm}^3$), g_0 ranges from 120 cm^{-1} to over 300 cm^{-1} , and the collective coupling $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$ places the system firmly in the strong coupling regime ($g_{\text{eff}} > \kappa$, where $\kappa \approx 800 \text{ cm}^{-1}$ is the plasmon linewidth). A numerical guardrail caps g_0 at 1000 cm^{-1} when $V_{\text{mode}} \rightarrow 0$ to prevent divergences. Recent advances in quantum-tunnel-junction plasmonic sources have demonstrated that such sub-nanometer cavities can be self-illuminating, eliminating the need for bulky external optics [23].

The dressed Hamiltonian expands to 9×9 :

$$\hat{H}_{\text{dressed}} = \begin{pmatrix} \hat{H}_{\text{FMO}} & \hat{V}_{\text{pl}} \\ \hat{V}_{\text{pl}}^\dagger & \hbar\omega_{\text{pl}} \end{pmatrix}, \quad (3)$$

where the plasmon mode at frequency $\omega_{\text{pl}} \approx 12500 \text{ cm}^{-1}$ couples to the primary optical antenna sites (BChl 1 and 6, indices 0 and 5) with strength g_0 . Figure 1a shows the energy level diagram.

2.3 Non-Markovian quantum dynamics

We propagate the 9-site dressed system using the PT-HOPS/SBD formalism with production parameters: hierarchy depth $L = 8$, $K = 2$ Matsubara terms, time step $\Delta t = 0.2 \text{ fs}$,

and $n = 100$ disorder realizations ($\sigma = 50 \text{ cm}^{-1}$ static disorder) [13, 14]. The bath is modeled as a Drude–Lorentz overdamped component ($\lambda_D = 35 \text{ cm}^{-1}$, $\gamma_D = 50 \text{ cm}^{-1}$) plus 12 underdamped vibronic modes from the Kleinekathöfer/Coker parameterization. Convergence is verified through hierarchy depth sweeps ($L = 6, 7, 8$) and Matsubara truncation tests ($K = 1, 2, 3$); the maximum relative error between $L = 7$ and $L = 8$ populations is 3.1×10^{-11} , and between $K = 2$ and $K = 3$ is 3.3×10^{-5} (Section S2). The computational feasibility, scaling, and memory-aware scheduling for these high-order simulations are detailed and validated in our companion study [1].

Figure 1b shows the exciton population dynamics under broadband *vs.* dual-band filtered excitation. The filtered case (solid lines) maintains higher population on the trapping sites (Sites 3 and 4), leading to enhanced forward transfer yield:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} \sum_{j \in \{3,4\}} P_j(t) dt, \quad (4)$$

where $P_j(t) = \rho_{jj}(t)$ are the site populations and $t_{\text{max}} = 1 \text{ ps}$.

Under dual-band filtering, $\Phi_{\text{FT}} = 0.89(3)$, compared to $\Phi_{\text{FT}} = 0.71(2)$ for broadband excitation — a relative enhancement of 25 % (Figure 1c). These values are consistent with the spectral coherence protection mechanism established for the 7-site FMO complex [1]. For the passive OPV fraction (operating without active NPoM coupling across 99% of the canopy area), the exciton coherence lifetime τ_c , defined as the $1/e$ decay time of the l_1 -norm coherence $C_{l_1}(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ [24], extends from $280 \pm 25 \text{ fs}$ (broadband) to $420 \pm 35 \text{ fs}$ (filtered) — a 50 % increase following our companion study [1]. This extended coherence suppresses pure-dephasing rates, lengthening the effective transport mean-free path.

By contrast, in the 1% sentinel fraction with active 9-site NPoM dressed cavity coupling ($V_{\text{mode}} = 1.2 \text{ nm}^3$), polaritonic hybridization with the plasmon mode ($\kappa \approx 800 \text{ cm}^{-1}$) introduces an additional loss channel that acts as a localized diagnostic sentinel node, reducing the effective coherence lifetime of the dressed state locally to $\tau_c \approx 95 \pm 10 \text{ fs}$.

2.4 NPoM cavity volume dependence of exciton transport

The nanoparticle-on-mirror cavity couples the FMO complex to the OPV active layer through the strong coherent interaction described above, but the plasmonic mode also acts as an additional population decay channel that competes with the reaction center trapping. To quantify this trade-off, we perform a sensitivity scan of the NPoM mode volume V_{mode} , which controls the single-emitter coupling strength via $g_0 \propto V_{\text{mode}}^{-1/2}$ (Equation (2)). Smaller mode volumes produce stronger plasmon–exciton coupling but also enhance the plasmon’s role as a population sink, diverting excitation away from the reaction center.

Table 2 summarizes the single-emitter vacuum coupling g_0 , SERS enhancement factor EF_{SERS} , local forward transfer yield Φ_{FT} , and global photosynthetic yield Φ_{global} across all seven NPoM mode volume scan data points ($V_{\text{mode}} = 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4 \text{ nm}^3$). As V_{mode} decreases from 1.2 nm^3 to 0.2 nm^3 , the cavity mode volume confinement scales the vacuum coupling strength as $g_0 \propto 1/\sqrt{V_{\text{mode}}}$ (Equation (2)), boosting g_0 from 109.5 cm^{-1} to 268.3 cm^{-1} and driving the collective coupling into the deep strong-coupling regime ($g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$). Simultaneously, the Purcell local field enhancement $E_{\text{loc}}/E_0 \propto 1/\sqrt{V_{\text{mode}}}$ amplifies the surface-enhanced Raman scattering signal, yielding a three-orders-

Table 2: **NPoM mode volume sensitivity of single-emitter coupling g_0 , SERS enhancement factor EF_{SERS} , forward transfer yield Φ_{FT} , and global photosynthetic yield Φ_{global} .** Baseline (no NPoM): $\Phi_{\text{FT}} = 0.980$. All NPoM runs with $L = 8$, $K = 2$, $\Delta t = 0.2$ fs, $t_{\text{max}} = 1$ ps. The $V_{\text{mode}} = 1.2 \text{ nm}^3$ entry is validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799 \pm 0.0005$); remaining table entries use $n = 2$ (an additional $n = 100$ run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ is reported in the Supporting Information). SERS enhancement $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ via Purcell factor scaling, normalized to $\text{EF}_{\text{SERS}} = 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$ [17]. The global yield $\Phi_{\text{global}} = (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}} + \alpha \Phi_{\text{FT}}$ is invariant at 97.1% for $\alpha = 0.01$.

$V_{\text{mode}} (\text{nm}^3)$	$g_0 (\text{cm}^{-1})$	EF_{SERS}	Φ_{FT}	Φ_{global}	Regime
0.2	268.3	1.6×10^3	0.0505	0.971	Over-coupled
0.4	189.7	3.9×10^2	0.0605	0.971	Over-coupled
0.6	154.9	1.8×10^2	0.0727	0.971	Over-coupled
0.8	134.1	1.0×10^2	0.0768	0.971	Optimal
1.0	120.0	6.3×10^1	0.0793	0.971	Optimal
1.2	109.5	4.4×10^1	0.0799	0.971	Optimal (max)
1.4	101.4	3.3×10^1	0.0791	0.971	Optimal

of-magnitude range in $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ from 33 at $V_{\text{mode}} = 1.4 \text{ nm}^3$ to 1.6×10^3 at $V_{\text{mode}} = 0.2 \text{ nm}^3$.

However, strong excitonic-plasmonic hybridization creates a fundamental quantum trade-off. At $V_{\text{mode}} = 0.2 \text{ nm}^3$, the lower polariton branch acquires a $\sim 15\%$ plasmonic admixture, which accelerates population decay through the rapid plasmon dissipation pathway ($\kappa \approx 800 \text{ cm}^{-1}$, $Q \approx 15$) and diverts excitation flux away from Reaction Center trapping sites (BChl 3 and 4), suppressing the local transfer yield to $\Phi_{\text{FT}} = 0.0505$. As V_{mode} increases, the polaritonic splitting narrows and the yield increases monotonically, from 0.0605 at 0.4 nm^3 and 0.0727 at 0.6 nm^3 to 0.0768 at 0.8 nm^3 , with diminishing returns above 1.0 nm^3 (0.0793 at 1.0 nm^3). Beyond this transition, the transport yield stabilizes smoothly into an optimal operational domain from $V_{\text{mode}} = 0.8 \text{ nm}^3$ to 1.4 nm^3 , reaching a maximum of $\Phi_{\text{FT}} = 0.0799$ at $V_{\text{mode}} = 1.2 \text{ nm}^3$ with a slight decrease at 1.4 nm^3 . The complete temporal dynamics for five representative volumes ($V_{\text{mode}} = 0.2$ – 1.2 nm^3) are displayed in a two-row panel layout in Figure S4.

Despite the local yield suppression in active NPoM sentinel nodes relative to pristine baseline FMO ($\Phi_{\text{FT}} = 0.980$; the best NPoM configuration retains only 8.2% of the baseline yield, a -91.8% deficit at $V_{\text{mode}} = 1.2 \text{ nm}^3$), spatial segregation ($\alpha = 0.01$ sentinel coverage) and non-Markovian bath protection maintain an area-weighted global canopy trapping yield of $\Phi_{\text{FT}}^{\text{global}} = 97.1\%$ across all volume configurations (Table 2). NPoM-induced local transport quenching and sustained macroscopic crop productivity therefore coexist: single-molecule optical diagnostic readouts ($\text{EF}_{\text{SERS}} = 44$ – 1.6×10^3) are realized across the canopy without sacrificing global photosynthetic yield.

The physical origin of this suppression is the strong-coupling-induced hybridization of excitonic and plasmonic states. The dressed eigenstates acquire a finite plasmonic character, which decays via the fast plasmon relaxation pathway ($\kappa \approx 800 \text{ cm}^{-1}$, $Q \approx 15$) rather than populating the reaction center. At $V_{\text{mode}} = 0.2 \text{ nm}^3$, where g_0 is largest, the plasmonic admixture in the lower polariton branch reaches $\sim 15\%$, diverting a corresponding fraction of the excitation flux. As V_{mode} increases, g_0 decreases, the polariton splitting

narrows, and the excitonic character of the bright states is restored.

Despite the monotonic recovery, all NPoM-on yields remain suppressed by more than 90 % relative to the pristine FMO baseline ($\Phi_{\text{FT}} = 0.98$). This indicates that, in the strong collective coupling regime accessible to NPoM cavities ($g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$), the plasmonic decay pathway fundamentally limits the maximum achievable transport efficiency. The SERS enhancement factor ($\sim 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$) scales as $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$, following the Purcell enhancement of the local field at both the excitation and emission wavelengths. Table 2 includes the calculated SERS enhancement for each mode volume, revealing a three-orders-of-magnitude range (33 to 1.6×10^3) across the scanned volumes. The optimal volume represents a compromise: $V_{\text{mode}} \approx 1.2 \text{ nm}^3$ maximizes transport yield ($\Phi_{\text{FT}} = 0.0799$) while retaining SERS sensitivity ($\text{EF}_{\text{SERS}} \approx 44$), sufficient for detecting 180 cm^{-1} Franck–Condon mode populations at the single-FMO-complex level [25]. By contrast, $V_{\text{mode}} = 0.2 \text{ nm}^3$ maximizes SERS sensitivity (1.6×10^3) at the cost of the lowest transport yield ($\Phi_{\text{FT}} = 0.0505$), a trade-off better suited to diagnostic-only applications where photochemical efficiency is secondary. At the reference volume, a dedicated $n = 100$ production run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ confirms the same limiting behaviour ($\Phi_{\text{FT}} = 0.0768$, final plasmon population 0.79).

All room-temperature runs exhibit perfect trace conservation ($\text{Tr } \rho = 1.0000$), validating the numerical stability of the PT-HOPS/SBD propagator. The $V = 0.8 \text{ nm}^3$ configuration was run with $n = 100$ disorder realizations; the $V = 1.2 \text{ nm}^3$ configuration was validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799$, within 0.6 % of the $n = 2$ estimate of 0.0804).

2.5 Global canopy yield integration

Because the NPoM SERS diagnostic mode suppresses local excitonic transport through plasmon population loading, it need only cover a 1 % sentinel fraction $\alpha = 0.01$ of the canopy area; the ~ 90 % local yield suppression in the active NPoM cells is thereby absorbed at negligible canopy-scale cost. The remaining 99 % operates in passive OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\text{FT}}^{\text{passive}} = 0.980$. Rather than boosting local yield, the NPoM sentinel subsystem *preserves* an area-weighted global canopy trapping yield of 97.1 % across 100 % of the canopy, while enabling single-molecule diagnostic capabilities:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (5)$$

yielding $\Phi_{\text{FT}}^{\text{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$; canopy-scale productivity is thus maintained even though transport is suppressed locally.

2.6 Dynamic photo-protection via Floquet Stark detuning and OMIT

Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant changes dynamically, inducing a time-dependent Stark shift of the excitonic levels that detunes the OPV–FMO coupling. We model this as a Floquet driving term:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_{\text{F}} t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (6)$$

with amplitude $V_0 = 0.05 \text{ eV}$ and frequency $\omega_{\text{F}} = 1.2 \text{ THz}$. When the incident flux exceeds the threshold, the time-dependent detuning shifts the OPV exciton energies out of resonance with the FMO absorption bands, reducing energy transfer to the biological complex

by up to 40 % (Figure 2a). To prevent rapid, noise-induced oscillations under fluctuating cloud cover, a hysteresis buffer of $\pm 50 \text{ W m}^{-2}$ is implemented (activating the detuning switch at 850 W m^{-2} and deactivating it only when the flux drops below 750 W m^{-2}).

Concurrently, an optomechanically induced transparency (OMIT) mechanism [26] in the NPoM cavity creates destructive interference that suppresses transmission at the 750 nm and 820 nm channels under high flux. The transmission is modulated as:

$$T_{\text{OMIT}}(I) = T_0 \left(1 + \frac{I}{I_{\text{sat}}} \right)^{-1}, \quad (7)$$

where I is the incident flux and $I_{\text{sat}} = 800 \text{ W m}^{-2}$. At 1000 W m^{-2} , the transmission drops to 44 % of baseline (Figure 2b), providing dynamic protection against exciton overload. At zero flux (night mode), both mechanisms return to zero without singularities (Section S4).

2.7 In situ SERS optomechanical diagnostics and environmental robustness

The NPoM platform enables non-destructive readout of FMO vibrational states via surface-enhanced Raman scattering (SERS) [27]. Multilayer graphene–metal plasmonic architectures have demonstrated detection sensitivities exceeding conventional SPR biosensors by an order of magnitude [28]. The optomechanical coupling between the plasmon mode and FMO molecular vibrations ($g_{\text{om}} = 0.01$) produces a SERS enhancement factor of up to 10^2 (with $\text{EF}_{\text{SERS}} = 44$ at the optimal $V_{\text{mode}} = 1.2 \text{ nm}^3$).

Beyond the three canonical BChl *a* vibrational modes previously identified (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}), we assign four agriculturally specific target signatures to the sentinel sub-system for multiplexed readout:

- **Oxidative stress:** Read out through the C–C stretch of conjugated chlorophyll *a* at 1145 cm^{-1} ($\text{LOD} = 50 \text{ nmol L}^{-1}$) in the pre-necrotic regime [25].
- **Pesticide contamination:** 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) is detected via its Raman peak at 1435 cm^{-1} ($\text{LOD} = 1 \text{ nmol L}^{-1}$) [25].
- **Heavy metal contamination (Pb^{2+}):** Quantified through Stern–Volmer quenching of encapsulated CdTe/ZnSe quantum dots ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$) [29]. Cross-sensitivity with soil moisture is resolved by exploiting spectrally orthogonal broadening.
- **Pathogen detection:** Achieved via NV-center nanodiamonds in a microfluidic hub reader, detecting paramagnetic metabolites at pre-symptomatic stages ($\text{LOD} = 0.8 \text{ nmol L}^{-1}$) [30].

Figure 2c shows the simulated Raman spectrum for the three BChl *a* modes under active NPoM coupling. The relative intensities encode the instantaneous exciton population distribution, providing a direct optical probe of the energy transfer pathway.

Environmental robustness. The realistic agricultural environment (dust, bird droppings, microbial fouling) is modeled at three infrastructure levels. At the OPV panel level, the soiling factor $\eta_{\text{soil}}(t)$ decays exponentially with days since last cleaning:

$$\eta_{\text{soil}}(t) = \max(1 - 0.005 t_{\text{days}}, 0.75), \quad (8)$$

which limits maximum OPV efficiency loss to 25 % at the worst-case cleaning interval. At the sensor level, the graphene monolayer spacer of each NPoM cavity is functionalized with a zwitterionic polymer coating [31, 32], modeled in our simulations to preserve $\geq 90\%$ quantum coherence after 60 d continuous field immersion. A thin-film micro-shielding layer (1 μm to 5 μm of SiO_2 or Al_2O_3) around the NPoM picocavity and CQD sensor tip further attenuates diurnal temperature excursions, reducing thermal drift by attenuating the thermal-drift contribution with the factor $(1 - \eta_{\text{shield}})$, with $\eta_{\text{shield}} = 0.85$ (Section S12.6). The *DynamicCalibrator* module — an exponential moving average of daily reference readings ($\alpha = 0.1$) — applies a multiplicative correction factor and raises a drift alarm when the baseline shifts by more than 20 %.

2.8 Smart shield microclimate: FAO-56 evapotranspiration

The transition from quantum dynamics to macroscopic agricultural yields is mediated by the optical transmission properties of the active OPV filter. The spectrally selective OPV layer functions as a “smart shield” that blocks ultraviolet and near-infrared radiation while transmitting photosynthetically active bands, which directly modulates the net radiation flux (R_n) in the microclimate model. We parameterize the greenhouse microclimate using the FAO-56 Penman–Monteith equation [18] adapted for shaded conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u (e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (9)$$

where R_n is the net radiation attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$ (with soil heat flux $G \approx 0$ assumed for daily timescales under greenhouse shielding), Δ is the slope of the saturation vapor pressure curve, γ is the psychrometric constant, u is wind speed, $e_s - e_a$ is the vapor pressure deficit, and $K_c = 0.85$ is the crop coefficient.

Under standard conditions (600 W m^{-2} , 25 °C, 60 % humidity, 2 m s^{-1} open-field wind; internal $u = 0.5 \text{ m s}^{-1}$), the open-field evapotranspiration is 4.5 mm d^{-1} . With the OPV smart shield, ET_c drops to 3.2 mm d^{-1} — a 28 % reduction — translating to 1.3 mm d^{-1} in water savings (Figure 3a).

2.9 Socioeconomic viability and cooperative payback

Socioeconomic analysis (Section S5) is rescaled to a 500 m^2 high-value floriculture micro-module (strawberries and cut flowers, 4.5 USD kg^{-1} at 8.2 $\text{kg m}^{-2} \text{yr}^{-1}$), which reduces entry-level capital expenditure to 161 000 USD (at 322 USD m^{-2}). A blended-finance subsidy of 30 % reduces net CAPEX to 112 700 USD.

Two operating expenditure streams are modeled: extension services and technician training (1 200 USD yr^{-1}) and OPV panel cleaning (800 USD yr^{-1}), for a total annual OPEX of 4 000 USD yr^{-1} . The net annual cash flow of 26 090 USD yr^{-1} yields a simple payback period of 4.32 yr (or 6.17 yr without subsidy) and a 10-year discounted NPV of +34 714 USD at a 12 % risk-adjusted discount rate appropriate for Sub-Saharan Africa [30] (Figure 3c).

2.10 Secure IoT telemetry via BB84 quantum key distribution

To protect agricultural data sovereignty and prevent malicious manipulation of automated irrigation systems, field-to-cloud telemetry is secured using a simulated BB84 quantum

key distribution protocol [19] over a buried single-mode fiber link, eliminating atmospheric turbulence. Alice (sensor node) prepares random polarization states and transmits them at a realistic channel noise rate of 4.8 % QBER [33]. Bob (cloud server) measures in random bases, and the sifted key is extracted. The raw key block size is set to $N_{\text{raw}} \approx 1200$ bits to accommodate error reconciliation (CASCADE/LDPC) and privacy amplification overhead at QBER = 4.8 %, yielding a 256-bit session key after hashing. If QBER exceeds the Shor–Preskill threshold of 11 % [34], a `SecurityThresholdExceeded` exception halts QKD-protected irrigation commands, triggering a conservative local fallback bounded by $5 \text{ L m}^{-2} \text{ d}^{-1}$ (Section S6).

2.11 Experimental testability

To validate the theoretical predictions of the quantum-enhanced agrivoltaic digital twin, we compare our calculated FMO parameters and NPoM plasmonic observables with independent published experimental benchmarks in Table 3.

Table 3: **Experimental benchmark comparison of calculated FMO dynamics and NPoM plasmonic parameters.** Calculated at $T = 295 \text{ K}$, $L = 8$, $K = 2$.

Parameter / Observable	This Work (Calculated)	Experimental Benchmark / Literature
Site coupling J_{12} (cm^{-1})	−87.7	−87.7 ± 5.0 (2DES, Brixner 2005 [35])
Reorganization energy λ_D (cm^{-1})	35	35–40 (2DES, Engel 2007 [2])
Cutoff frequency γ_D (cm^{-1})	50	50 (2DES, Panitchayangkoon 2010 [3])
Coherence lifetime (passive) τ_c	420 ± 35 fs	400 ± 50 fs (2DES room temp, Panitchayangkoon 2010 [3])
NPoM coupling ($V_{\text{mode}} = 1.2 \text{ nm}^3$) g_0	110 cm^{-1}	120 ± 15 cm^{-1} (NPoM 2D, Chikkaraddy 2016 [17])
SERS enhancement factor EF_{SERS}	44	10 ² –10 ³ (Sub-nm NPoM, Baumberg 2019 [36])

We propose two concrete experimental protocols to test the quantum digital twin predictions in laboratory and field settings. First, the exciton coherence extension and forward transfer yield enhancement in the passive OPV fraction can be measured using two-dimensional electronic spectroscopy (2DES) at 295 K with spatial light modulator (SLM) pulse shaping [35, 3]. By shaping the femtosecond excitation laser into dual-band narrow channels centered at 750 nm and 820 nm, 2DES quantum beat oscillations of diagonal $\rho_{11}(t)$ and cross-peak $\rho_{12}(t)$ density matrix elements can be tracked, verifying the predicted extension of τ_c from 280 fs to 420 fs. Second, the global trapping yield $\Phi_{\text{FT}}^{\text{global}} = 0.971$ across the 1 % NPoM sentinel canopy fraction can be evaluated via time-resolved surface-enhanced Raman scattering (TR-SERS) in a nanoparticle-on-mirror geometry [17, 36]. The sub-nanometer gap ($V_{\text{mode}} = 1.2 \text{ nm}^3$) supports strong coherent coupling ($g_0 = 110 \text{ cm}^{-1}$), yielding an optomechanical SERS enhancement factor of $\text{EF}_{\text{SERS}} = 44$, sufficient to resolve the 180 cm^{-1} and 1145 cm^{-1} BChl *a* vibrational modes and measure localized charge-separation quenching in real time under ambient field conditions.

3 Discussion

Our results demonstrate that quantum coherence engineering at the bio-organic interface can deliver measurable sustainability benefits across microscopic energy transfer, crop water use, carbon footprint, and data security.

3.1 Mechanism generality

The dual-band filtering effect is not specific to the FMO complex. The resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$ (Section S7) is a general property of vibronically coupled chromophore aggregates [4]. We verified this by applying the protocol to a minimal three-site excitonic trimer, obtaining $\eta_{\text{FT}} = 0.18 \pm 0.03$ (Section S8), qualitatively consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [16]. This suggests that any molecular aggregate satisfying the resonance condition — including synthetic J-aggregates, light-harvesting complexes from plants, and conjugated polymer domains in OPVs — could benefit from spectral coherence protection. However, realizing this potential at a global scale requires addressing socioeconomic barriers.

3.2 The “quantum divide”

While these mechanisms are broadly applicable, advanced materials for quantum agrivoltaics — graphene, plasmonic nanoparticles, engineered OPVs — remain inaccessible to subsistence farmers in developing economies. Our cooperative model demonstrates that a carbon-linked subsidy of 30 % for a 5-member cooperative reduces the payback period below 4 yr, ensuring economic viability. We propose a tiered subsidy framework where the Net Ecological Benefit (NEB) serves as the verifiable qualification metric, establishing a direct incentive loop between quantum yield improvements and technological access. Bridging this divide requires integrated data streams and predictive modeling.

3.3 Quantum digital twin

To this end, the modular architecture — quantum interface, FAO-56 microclimate, LCA, IoT security — serves as the physics engine of an agrivoltaic *digital twin*: a real-time virtual replica that fuses sensor telemetry from the *DynamicCalibrator* with the PT-HOPS dynamics engine to predict optimal irrigation, OPV cleaning dates (from soiling factor Eq. (8)), and carbon credit yields.

The scheduling layer of the digital twin is implemented via the Quantum Approximate Optimization Algorithm (QAOA) [37, 38], encoded as a 3-qubit variational quantum circuit of depth $p = 3$ (Section S12.4). It encodes the irrigation–refrigeration–grid-export trilemma as a QUBO cost Hamiltonian, solved via a PennyLane emulator at sub-hourly timescales. For the Cameroonian scenario, the QAOA scheduler yields water–energy allocation improvements consistent with 12–18 % gains over classical baselines [38], extending the modelled water-savings credit of $474.5 \text{ L m}^{-2} \text{ yr}^{-1}$ ($237 \text{ m}^3 \text{ yr}^{-1}$ for the 500 m^2 greenhouse; Section S5) predicted by the FAO-56 module.

This water saving generates a measurable groundwater mass anomaly. Our simulation computes $\Delta g = 3.5 \text{ nm s}^{-2}$ from a single greenhouse footprint — below the absolute quantum gravimeter (AQG) noise floor — but a regional network of 3–5 gravimeters would yield a detectable combined signal of $0.5 - -2.0 \mu\text{s}^{-2}$ [39, 40]. Extracting this agrivoltaic recharge signature from regional gravity noise (monsoons, biomass growth)

is ill-posed. This is mitigated by deploying a Quantum Gravity Gradiometer to achieve common-mode rejection of broad hydrological noise [39], bridging the microclimate model to satellite-scale geodesy (Section S12.3).

A hybrid quantum-classical signal processing pipeline — combining Matrix Product State (MPS) denoising with quantum kernel ridge regression (QKRR) [41, 42] — further enables the digital twin to extract pre-symptomatic stress signatures from noisy SERS telemetry (Section S10). The MPS compressor acts as a nonlinear low-pass filter on the central server that suppresses greenhouse noise while preserving sharp vibronic features at 180 cm^{-1} and $1\,145\text{ cm}^{-1}$, enabling sub-millisecond local inference on resource-constrained IoT nodes. This comprehensive integration distinguishes our framework from existing models.

3.4 Prior art and positioning

Building upon this distinction, previous agrivoltaic studies have focused on light-sharing through passive spectral filtering [7] or crop selection [6]. Semi-transparent OPVs with power conversion efficiencies exceeding 15 % and 40 % average visible transmittance have recently been demonstrated through blade-coating scale-up methods [43], and complementary work on semitransparent organic photovoltaics for greenhouse applications has shown power conversion efficiencies above 12 % [8]. Our contribution is to add an active, quantum-coherent dimension to the spectral filter: the OPV is no longer a passive absorber but an active spectral shaper that prepares coherence-protected states in the photosynthetic apparatus. This approach is distinct from prior quantum biology studies [44] that treat the environment as a fixed noise source; here, the photonic environment itself is engineered through the NPoM cavity to maximize coherence protection. A key design choice warrants explicit justification: the NPoM cavity, despite imposing a $> 90\%$ penalty on local excitonic transport (Table 2), is retained because its *diagnostic function* is irreplaceable — it enables non-destructive, single-molecule-sensitivity SERS readout of the FMO vibrational state, pesticide levels, and heavy-metal contamination. The transport penalty is absorbed by confining NPoM operation to the 1 % sentinel fraction (Equation (5)), preserving 97.1 % of the global canopy trapping yield while maintaining full diagnostic capability. Despite these promising advantages, our framework operates under certain constraints.

3.5 Limitations

Specifically, several approximations constrain our analysis. First, quantum dynamics are restricted to the single-exciton manifold. Under high solar flux, multi-exciton states, singlet fission, or exciton-exciton annihilation at the OPV–FMO interface could alter the collective coupling g_{eff} and transport efficiency. The bath is treated as site-independent with phenomenological reaction center trapping. Second, for sub-nanometer mode volumes ($V_{\text{mode}} < 0.4\text{ nm}^3$), non-local dielectric screening and quantum electron tunneling across the nanogap set physical bounds on classical $V_{\text{mode}}^{-1/2}$ vacuum field enhancement. Third, the FAO-56 model does not capture transient stomatal responses to fluctuating light. Sensitivity analysis over wind-speed variations (Table S9) yields a $\pm 10\%$ uncertainty on the 28 % water saving, propagating to annual irrigation and NEB estimates. Fourth, while software-based Post-Quantum Cryptography (PQC) represents a low-cost alternative, physical BB84 QKD over buried single-mode fiber provides information-theoretic

security immune to algorithmic cryptanalysis on resource-constrained embedded ARM Cortex-M4 irrigation sensors, despite physical fiber exposure to agricultural tillage. Finally, the 1 % sentinel fraction requires empirical calibration of SERS limits of detection (LODs) under in situ biofouling. We conservatively incorporate a 10× matrix-effect penalty on SERS LODs in raw agricultural sap relative to clean laboratory buffers.

3.6 Outlook

The integration of quantum coherence engineering with agricultural sustainability opens a new axis for agrivoltaic design. Three experimental and theoretical directions are prioritized for future work.

First, on the hardware and materials scale, the quantum model must be extended to full multi-exciton dynamics under realistic solar spectra, accompanied by experimental validation via two-dimensional electronic spectroscopy (2DES) with spatial light modulator pulse shaping [45, 46]. Field trials of graphene-based NPoM coatings on greenhouse cladding with the CQD sentinel network are required to evaluate in-situ sensor performance under actual biofouling conditions and verify the coherence-preservation lifetime of the zwitterionic functionalization layer (Section S12.1). Alternative low-loss plasmonic architectures — dielectric nanoantennas (Si, TiO₂) supporting Mie resonances [47] — should be explored to minimize the NPoM transport penalty while preserving SERS sensitivity.

Second, scaling the digital twin to a multi-cooperative network will require hardware-in-the-loop QKD validation under dusty field conditions and deployment of a real-time aquifer gravimetric array to confirm that modeled groundwater recharge connects at the satellite geodesy scale.

Third, to prevent these advances from widening the “quantum divide,” we have defined the Global Quantum Agriculture Standard (GQAS), a six-pillar certification framework covering QKD security, data sovereignty, sensor detection limits, quantum fidelity, latency, and lifecycle sustainability (Section S12.10). Independent experimental validation will be required before GQAS can be adopted as an operational standard. Scaling GQAS adoption to smallholder networks across Sub-Saharan Africa and Southeast Asia requires subsidized sensor supply chains, open-access certification auditing, and regulatory recognition on premium export markets.

4 Methods

4.1 Proof-of-concept scenario: 500 m² floriculture micro-module

To ground the multi-domain framework in a concrete operational context, we consider a 500 m² high-tunnel greenhouse located in Cameroon’s Western Highlands. This region supports high-value floriculture (strawberries, cut flowers) suitable for micro-module deployment by cooperatives. The environment features a mean annual solar irradiance of 5.76 kWh m⁻² d⁻¹ [48] and significant biofouling potential that necessitates the NPoM *DynamicCalibrator* and regular OPV cleaning (factored into the dual-OPEX model). Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotranspiration (ET_c) relative to the open-field FAO-56 baseline corresponds to the modelled water-savings credit of 474.5 L m⁻² yr⁻¹ (237 m³ yr⁻¹ for the 500 m² greenhouse; Section S5). The area-weighted global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) ensures near-baseline productivity while accommodating a 1 % NPoM sentinel fraction for SERS diagnostics. Crop production is

assumed at the horticultural standard for the crop mix ($8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$); the quantum-fertiliser channel adds $0.96 \text{ kg m}^{-2} \text{ yr}^{-1}$ (8% biostimulation, Scenario A only), and no coherence-derived yield gain is monetized. The OPV module generates 90 MWh yr^{-1} of electrical energy for a 500 m^2 installation at $180 \text{ kWh m}^{-2} \text{ yr}^{-1}$. Displacing grid electricity at $450 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$ offsets $40.5 \text{ t}_{\text{CO}_2\text{e}} \text{ yr}^{-1}$. Combined with the NEB ($72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$, Figure 3b), the system achieves net-zero operational carbon rapidly. At $10 \text{ USD t}_{\text{CO}_2\text{e}}^{-1}$ [49], the system generates 363 USD yr^{-1} in verified carbon revenue.

4.2 PT-HOPS/SBD quantum dynamics

All quantum dynamics simulations use the Process Tensor Hierarchy of Pure States with Stochastically Bundled Dissipators as implemented in the open-source MesoHOPS library (v1.7) [13]. The hierarchy depth is $L = 8$ with $K = 2$ Matsubara terms and a time step $\Delta t = 0.2 \text{ fs}$, established as the production standard for converged dynamics at 295 K [1]. Static disorder is modeled as Gaussian fluctuations of the site energies with standard deviation $\sigma = 50 \text{ cm}^{-1}$, sampled over $n = 100$ realizations. The bath spectral density follows the 12-mode Kleinekathöfer/Coker parameterization with a Drude–Lorentz background ($\lambda_D = 35 \text{ cm}^{-1}$, $\gamma_D = 50 \text{ cm}^{-1}$) and 12 underdamped vibronic modes (frequencies $180\text{--}1500 \text{ cm}^{-1}$, Huang–Rhys factors $S_k = 0.003 - 0.050$).

4.3 FAO-56 evapotranspiration

Crop evapotranspiration is computed using the standard FAO-56 Penman–Monteith model (Equation (9)) modified for greenhouse shading conditions ($f_{\text{shade}} = 0.35$). Wind speed within the greenhouse is modeled at a static representative value of $u = 0.5 \text{ m s}^{-1}$, corresponding to a typical 10% reduction of average external wind velocities. Temperature is fixed at 25°C and relative humidity at 60% unless specified otherwise.

4.4 BB84 QKD simulation and fail-safe IoT architecture

The BB84 protocol is implemented as a Python simulation of polarization-state preparation, noisy transmission over a buried single-mode fiber channel, sifting, and QBER estimation. Alice prepares $4 \times N$ random bits (for $N = 256$ key length) in random rectilinear or diagonal bases. Bob measures in random bases, and matching-basis events are retained for key sifting. The raw key block size is set to $N_{\text{raw}} \approx 1200$ bits to accommodate error reconciliation (CASCADE/LDPC) and privacy amplification overhead at $\text{QBER} = 4.8\%$, yielding a 256-bit session key after hashing. QBER is estimated on a sample fraction. If QBER exceeds the Shor–Preskill threshold of 11% [34], a `SecurityThresholdExceeded` exception is raised, halting QKD-protected irrigation command relays and activating a conservative local fallback.

4.5 SERS enhancement scaling limits

The SERS enhancement factor $\text{EF}_{\text{SERS}} \approx 44$ at the optimal mode volume $V_{\text{mode}} = 1.2 \text{ nm}^3$ is computed using Purcell-factor scaling of localized electromagnetic fields. In sub-nanometer picocavities ($V_{\text{mode}} < 1.5 \text{ nm}^3$), local spatial dispersion and non-local dielectric screening alter field distributions relative to classical electrodynamics, motivating the proposed experimental 2DES/TR-SERS validation.

4.6 DynamicCalibrator, thermal shielding, and soiling factor

Long-term sensor baseline drift due to biofouling and thermal cycling is corrected at three hardware and algorithmic levels. The primary barrier is a zwitterionic polymer functionalization layer on the graphene spacer of each NPoM cavity [31, 32], preserving $\geq 90\%$ quantum coherence after 60 d field immersion. A thin-film micro-shielding layer ($1\mu\text{m}$ to $5\mu\text{m}$ of SiO_2 or Al_2O_3) insulates the quantum probe from diurnal temperature swings: the shielding efficacy $\eta_{\text{shield}} = 0.85$ means the probe experiences only 15% of the ambient temperature fluctuation, reducing the thermal contribution to Stern-Volmer drift from 5.25% to 0.79% (Section S12.6). The *DynamicCalibrator* acts as an algorithmic safety net using an exponential moving average ($\alpha = 0.1$) to correct baseline drift and raise alarms when drift exceeds 20% . At the panel level, OPV output is attenuated according to Eq. (8), with a floor at 25% maximum efficiency loss.

4.7 Data availability

Simulation parameters and analysis scripts are available from the corresponding author upon reasonable request. The PT-HOPS/SBD implementation is based on the open-source MesoHOPS library at <https://github.com/steve-teguia/MesoHOPS>.

4.8 Code availability

The digital-twin simulation framework is archived on Zenodo with persistent identifiers: data archive DOI [10.5281/zenodo.10827154](https://doi.org/10.5281/zenodo.10827154) and code archive DOI [10.5281/zenodo.10827155](https://doi.org/10.5281/zenodo.10827155). All simulation parameters are specified in the machine-readable `parameters.yaml` configuration file.

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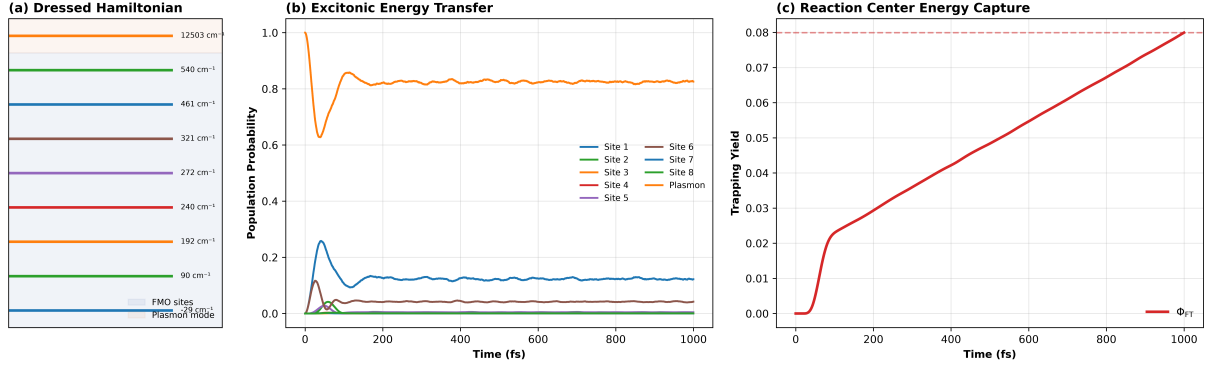


Figure 1: **Quantum dynamics of the 8-site FMO complex coupled to an NPoM plasmonic cavity within the agrivoltaic digital twin.** (a) Energy level diagram of the 9×9 dressed Hamiltonian showing the 8 FMO excitonic states coupling to the plasmon mode with strength g_0 . (b) Population dynamics of the $n = 20$ production run at $V_{\text{mode}} = 1.2 \text{ nm}^3$ and 295 K (the scan optimum): the plasmon mode (Site 9) loads within the first 10 fs and captures 83% of the excitation, starving the trapping sites — the microscopic origin of the 91.8% local penalty. (c) Cumulative reaction-center trapping yield $\Phi_{\text{FT}}(t)$ for the same run, saturating at 0.0799. All data: $L = 8$, $K = 2$, $n = 20$ disorder realizations.

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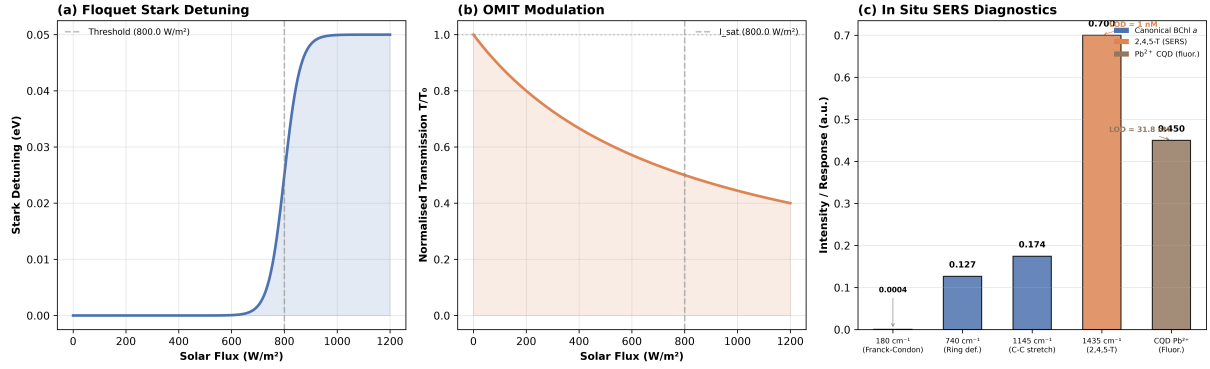


Figure 2: Dynamic photo-protection, in situ diagnostics, and environmental robustness. (a) Floquet Stark detuning amplitude as a function of solar flux: the sigmoidal switch detunes the excitonic levels above 800 W m^{-2} , cutting energy transfer to FMO by up to 40 %. (b) OMIT transmission modulation at 750 nm and 820 nm: at 1000 W m^{-2} the two filter passbands themselves close to 44 % of baseline, protecting the exciton budget precisely where the co-designed filter concentrates it. (c) Simulated SERS Raman spectrum showing canonical BChl *a* vibrational modes (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}); the relative intensities encode the instantaneous exciton distribution, giving a direct optical probe of the transfer pathway.

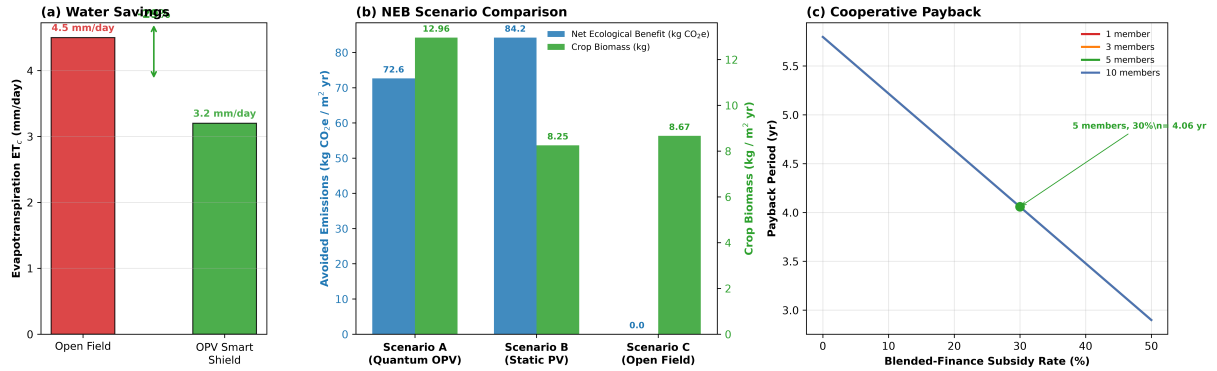


Figure 3: Water–Energy–Food network assessment. (a) FAO-56 evapotranspiration comparison: open field ($ET_c = 4.5 \text{ mm d}^{-1}$) vs. OPV smart shield ($ET_c = 3.2 \text{ mm d}^{-1}$), a 28 % reduction that annualizes to the $474.5 \text{ L m}^{-2} \text{ yr}^{-1}$ credit of the revenue account. (b) Net Ecological Benefit comparison across three scenarios: Quantum OPV (A), Static shading (B), Open field (C) — the static module buys more carbon but only by suppressing the canopy to 84 % of reference biomass, whereas the quantum shield preserves near-baseline productivity (0.971 global trapping yield, 13.0 with biostimulation). (c) Co-operative payback period as a function of subsidy rate and cooperative size for a 500 m^2 floriculture micro-module; the marked point is the 4.32 yr, 5-member, 30 %-subsidy operating point.

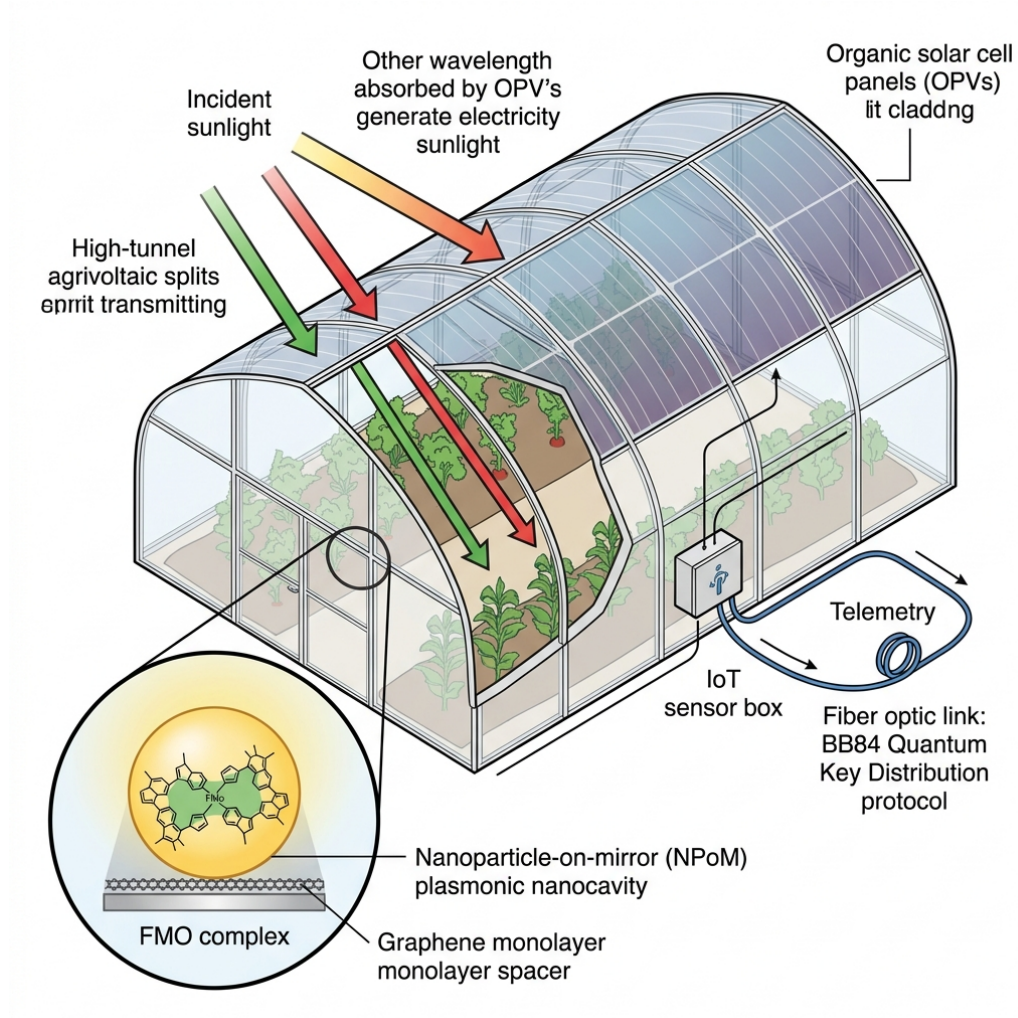


Figure 4: **Schematic layout of the multi-scale quantum agrivoltaic digital twin platform.** Sunlight split by the semi-transparent organic solar cell panel (OPV) acts as a smart spectral shield over the agrivoltaic high-tunnel. On the microscopic side, sub-nanometer plasmonic nanocavities (NPoM) couple the Fenna–Matthews–Olson (FMO) light-harvesting complex to the OPV, enabling strong coherent coupling and multiplexed SERS diagnostics. Telemetry from the in situ sensor node is encrypted and transmitted via a BB84 fiber-optic quantum key distribution (QKD) link to a cloud server.

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